

Deploy an automated build pipeline using dockerhub

Step 1: Create GitHub Personal Access Token (PAT)

Go to GitHub

Create a token

Give it permission to write packages

Save it safely

Step 2: Build the Docker image

Run: `docker build -t program-6 .`

```

lrw24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pgm6$ docker build -t prog6 .
[+] Building 11.2s (2/3)                                docker:default
=> [internal] load build definition from Dockerfile      0.0s
=> => transferring dockerfile: 201B                    [+] Building 11.4
or docker.io/library/python:3.9-slim                    => [internal] load build definition from Dockerfile
0 => => transferring dockerfile: 201B
ult] load metadata for docker.io/library/python:3.9-slim => [internal] load build definition from Dockerfile
registry-1.docker.io => => transferring dockerfile: 201B
docker:default] load metadata for docker.io/library/python:3.9-slim => [internal] load build definition from Dockerfile
python:pull token for registry-1.docker.io => => transferring dockerfile: 201B
docker:default] load metadata for docker.io/library/python:3.9-slim
0.0sbrary/python:pull token for registry-1.docker.io => => transferring dockerfile: 201B
docker:default] load metadata for docker.io/library/python:3.9-slim
0.0sbrary/python:pull token for registry-1.docker.io
ng 12.1s (2/3)                                docker:default] load metadata for docker.io/library/python:3.9-slim
m Dockerfile                                0.0sbrary/python:pull token for registry-1.docker.io
[+] Building 12.3s (2/3)                                docker:default
build definition from Dockerfile                    0.0sbrary/python:pull token
[+] Building 12.4s (2/3)
=> [internal] load build definition from Dockerfile      0.0sbrary/python:pull token
: 201B                                [+] Building 12.6s (2/3)
9-slim => [internal] load build definition from Dockerfile
nsferring dockerfile: 201B                                [+] Building 12.7s (2/3)
io/library/python:3.9-slim => [internal] load build definition from Dockerfile
=> => transferring dockerfile: 201B                                [+] Building 12.8s (2/3)
metadata for docker.io/library/python:3.9-slim => [internal] load build definition from Dockerfile
1.docker.io => => transferring dockerfile: 201B
ocker:default] load metadata for docker.io/library/python:3.9-slim => [internal] load build definition from Dockerfile
docker:default => [internal] load build definition from Dockerfile
0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 14.5s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 467B 0.0s
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f1 0.0s
=> CACHED [2/5] WORKDIR /app 0.0s
=> CACHED [3/5] COPY requirements.txt . 0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt 0.0s
=> [5/5] COPY . . 0.0s
=> exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:a8a90d1075a6aa0e1eb89907c4a2a261d39cb371e9cec8a09366dcfd7e7 0.0s

```

Step 3: Tag the image (Rename for Docker Hub)

```
docker tag program-6 <your-dockerhub-name>/program-6:latest
```

```
1rv24mc055 kishan s@kishan-L00-15IAX9:~/lab/pgm6$ docker tag prog6 kishankharvi/prog6:latest
```

Step 4: Push image to Docker Hub

First login:

```
docker login -u <dockerhub-username>
```

Then push:

```
docker push <dockerhub-username>/program-6:latest
```

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pgm6$ docker push kishankharvi/prgm6
Using default tag: latest
The push refers to repository [docker.io/kishankharvi/prgm6]
f21146b4ef01: Pushed
f222c41a8915: Mounted from kishankharvi/program6
c4e0be102a55: Mounted from kishankharvi/program6
2455e828139e: Mounted from kishankharvi/program6
c8f6b54339a8: Mounted from kishankharvi/program6
298992e09a03: Mounted from kishankharvi/program6
4f237755fbae: Mounted from kishankharvi/program6
d7c97cb6f1fe: Mounted from kishankharvi/program6
latest: digest: sha256:f66d7565d9c4eadb551d842231783247dee0b7ddd49017e53916c6018ded70b6 size: 1990
```

Step 5: Tag the image for GitHub Container Registry (GHCR)

```
docker tag program-6 ghcr.io/<github-username>/program-6:latest
```

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pgm6$ docker tag prgm6 ghcr.io/kishankharvi/prgm6:latest
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pgm6$ export GITHUB_PAT=ghp_Kan44GQca4TPSqdPDz3PP7gYUrau0P1qTbQX
```

Step 6: Push to GHCR

Login using your token:

```
echo $GITHUB_PAT | docker login ghcr.io -u <username> --password-stdin
```

```
Push:
docker push ghcr.io/<github-username>/program-6:latest
```

Step 7: Connect your local project to GitHub

If your local folder isn't linked to GitHub yet:

```
git remote add origin https://github.com/<username>/program-6.git
```

This tells Git:

"Whenever I push code → send it to this GitHub repo."

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pgm6$ echo $GITHUB_PAT | docker login ghcr.io -u kishankharvi --password-stdin
```

```
WARNING! Your credentials are stored unencrypted in '/home/1rv24mc055_kishan_s/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/
```

```
Login Succeeded
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pgm6$ docker push ghcr.io/kishankharvi/prgm6:latest
The push refers to repository [ghcr.io/kishankharvi/prgm6]
f21146b4ef01: Pushed
f222c41a8915: Mounted from kishankharvi/program6
c4e0be102a55: Mounted from kishankharvi/program6
2455e828139e: Mounted from kishankharvi/program6
c8f6b54339a8: Mounted from kishankharvi/program6
298992e09a03: Mounted from kishankharvi/program6
4f237755fbae: Mounted from kishankharvi/program6
d7c97cb6f1fe: Mounted from kishankharvi/program6
latest: digest: sha256:51e1481381e64e4f7161a71a312786ffdea670e191f5f60e6290350667462d1e size: 1990
```

Step 8: Push your source code to GitHub

```
git add .
```

```
git commit -m "Initial commit"
```

```
git push -u origin main
```

Step 9: Check both registries

- On GitHub = check Packages
- On Docker Hub = check Repositories

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pgm6$ git branch -M main
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pgm6$ git remote add origin https://github.com/Kishankharvi/prgm6.git
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pgm6$ git add .
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pgm6$ git commit -m "initial commit"
[main (root-commit) 2b7cbef] initial commit
3 files changed, 31 insertions(+)
create mode 100644 Dockerfile
create mode 100644 app.py
create mode 100644 requirements.txt
-----
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pgm6$ git push -u origin main
Username for 'https://github.com': kishankharvi
Password for 'https://kishankharvi@github.com':

Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 574 bytes | 574.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Kishankharvi/prgm6.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```